```
Authorization: Basic ZGVtbzpjaGFuZ2VpdA==
```

The Authorization header is a standard HTTP header for sending credentials to the server. It's extensible, allowing different authentication schemes[3], but in this case you're using the Basic scheme. The credentials follow the authentication scheme identifier. For Basic authentication, these consist of a string of the username followed by a colon[4] and then the password. The string is then converted into bytes (usually in UTF-8, but the standard does not specify) and Base64-encoded, which you can see if you decode it in jshell:

```
jshell> new String(
java.util.Base64.getDecoder().decode("ZGVtbzpjaGFuZ2VpdA=="),
"UTF-8")
$3 ==> "demo:changeit"
```

> **WARNING** HTTP Basic credentials are easy to decode for anybody able to read network messages between the client and the server. You should only ever send passwords over an encrypted connection. You'll add encryption to the API communications in section 3.5.

Base64 encoding

Base64 is a way of encoding binary data into a text format using only printable ASCII characters. The way it works is to treat the input as a sequence of binary bits, split into 6-bit chunks. Each chunk is then treated as a number between 0 and 63 and then encoded using an alphabet of 64 symbols. Just as decimal uses the symbols 0 to 9 to represent the 10 possible digits, and hexadecimal uses digits 0 to 9 and characters A to F to represent the 16 hex digits, so the Base64 encoding scheme uses 64 symbols to encode 6-bit values.